Design and Analysis of Algorithm

Course Code: MSCCS24101

January 13, 2026

Contents

1 Course Overview

Course Code: MSCCS24101
Credits: 0-0-0

2 Course Outcomes (COs)

CO	Description	Bloom's Level
CO1	Analyze worst-case running times of algorithms using asymptotic analysis.	Analyze
CO2	Compare between different data structures. Pick an appropriate data structure for a design situation.	Understand
CO3	Ability to design algorithms using standard paradigms like: Greedy, Divide and Conquer, Dynamic	Create
CO4	Able to Explain the major graph algorithms and Employ graphs to model engineering problems, when	Understand
CO5	Able to Compare between different data structures and pick an appropriate data structure for a design	Understand

3 Course Content

3.1 Unit I: Introduction to Algorithms (10 Hours)

- Algorithm definition and characteristics, Space complexity, Time complexity- worst case, best 1 1,3
- case, average case, Complexity, asymptotic notation, Recursive and non-recursive algorithms,
- Sorting algorithms : insertion sort, heap sort, bubble sort, Sorting in linear time: counting sort,
- Searching algorithms: Linear, Binary

3.2 Unit II: Divide and Conquer strategy and Greedy (14 Hours)

- General method, control abstraction, Binary search, Merge sort, Quick sort, Comparison
- between Traditional Method of Matrix Multiplication vs. Strassen"s Matrix Multiplication,
- Writing simple algorithm using Divide and conquer strategy: power(x,n), find occurrence of a
- number from array of N integers, to find minimum from an array, mini-max algorithm, largest
- number multiplication, simple convex algorithm

- Greedy Method: Knapsack problem, Job sequencing with deadlines, Minimum-cost spanning
- trees: Kruskal and Prim"s algorithm, Optimal storage on tapes, Optimal merge patterns,
- Huffman coding, Shortest Path :Dijkstra"s Algorithm

3.3 Unit III: Dynamic Programming and Decrease and (16 Hours)

- Principle of optimality, Matrix chain multiplication, 0/1 Knapsack Problem i)Merge & Purge
- ii)Functional Method, Bellman Ford Algorithm, All pairs Shortest Path Floyd-Warshall
- Algorithm, Travelling Salesperson problem, Longest common subsequence, String editing
- Decrease and conquer: Definition of Graph Representation of Graph, By Constant - DFS and
- BFS, Topological sorting, Strongly Connected components and spanning trees, Articulation
- Point and Bridge edge

3.4 Unit IV: Backtracking, Branch and Bound (16 Hours)

- General method, Fixed Tuple vs. Variable Tuple Formulation, n- Queen"s problem, Graph
- Branch and Bound Techniques: Introduction : Branch and bound terms like definition of live
- node, E-node, Dead node, Least cost (LC) search, Least cost Branch and Bound (LCBB)
- 0/1 knapsack problem using LCBB method (fixed tuple size), Travelling Salesman problem
- using LCBB method (variable tuple size)

3.5 Unit V: Problem Classification (4 Hours)

- The class of P, NP, NP-hard and NP -Complete, Relationship among P
- class, NP class, NP-hard and NP -Complete, Cook"s theorem

4 CO-PO Mapping Matrix

CO	Strongest PO	Affinity	Rationale
CO1	PO1	3	Analysis

5 References

1. 2 Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, Pearson Education, Reprint 2006. 3 Algorithms Design and Analysis, Harsh Bhasin, Oxford university press, 2016. 4 Design and Analysis of Algorithms, S. Sridhar, Oxford university press, 2014. M.Sc - Computer sci

6 Optimization Analysis

6.1 Bloom's Taxonomy Distribution

Level	**Count**
Analyze	1
Understand	3
Create	1